



Multi-scale feature fusion for prediction of IDH1 mutations in glioma histopathological images

Xiang Liu^{a,b,1}, Wanming Hu^{c,1}, Songhui Diao^{a,b,1}, Deboch Eyob Abera^{a,b}, Daniel Racocanu^d, Wenjian Qin^{a,*}

^a Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China

^b Shenzhen College of Advanced Technology, University of Chinese Academy of Sciences, Shenzhen 518055, China

^c Department of Pathology, Sun Yat-sen University Cancer Center, State Key Laboratory of Oncology in South China, Guangdong Provincial Clinical Research Center for Cancer, Guangzhou 510060, China

^d Sorbonne University, Inria, CNRS, Inserm, AP-HP, Inria, Paris Brain Institute - ICM, F-75013 Paris, France

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ABSTRACT

Background and objective: Mutations in isocitrate dehydrogenase 1 (IDH1) play a crucial role in the prognosis, diagnosis, and treatment of gliomas. However, current methods for determining its mutation status, such as immunohistochemistry and gene sequencing, are difficult to implement widely in routine clinical diagnosis. Recent studies have shown that using deep learning methods based on pathological images of glioma can predict the mutation status of the IDH1 gene. However, our research focuses on utilizing multi-scale information in pathological images to improve the accuracy of predicting IDH1 gene mutations, thereby providing an accurate and cost-effective prediction method for routine clinical diagnosis.

Methods: In this paper, we propose a multi-scale fusion gene identification network (MultiGeneNet). The network first uses two feature extractors to obtain feature maps at different scale images, and then by employing a bilinear pooling layer based on Hadamard product to realize the fusion of multi-scale features. Through fully exploiting the complementarity among features at different scales, we are able to obtain a more comprehensive and rich representation of multi-scale features.

Results: Based on the Hematoxylin and Eosin stained pathological section dataset of 296 patients, our method achieved an accuracy of 83.575 % and an AUC of 0.886, thus significantly outperforming other single-scale methods.

Conclusions: Our method can be deployed in medical aid systems at very low cost, serving as a diagnostic or prognostic tool for glioma patients in medically underserved areas.

1. Introduction

Brain tumors have attracted widespread attention due to their high morbidity and mortality. Among them, glioma is the most common type of primary brain tumor [1]. Indeed, gliomas account for almost 30 % of primary brain tumors and 80 % of malignant tumors, and are the leading cause of death from primary brain tumors [2]. Genome-wide mutation analysis of gliomas has shown that mutations in IDH1 and IDH2 exist in a proportion of gliomas, which holds immense clinical value in the treatment of glioma [3], and patients with IDH mutations have longer overall survival compared to those without the mutation [4]. IDH is an

important prognostic, diagnostic, and therapeutic biomarker for glioma. This led to the comprehensive genome-histological characteristics of glioma, proposed in the revised 4th Edition World Health Organization (WHO) classification of tumours of the central nervous system in 2016 [5]. In addition, the role of IDH was further strengthened in the 5th edition of the WHO Central Nervous System Classification in 2021 [6]. According to the status of IDH mutation, adult gliomas can be divided into three categories: astrocytoma IDH mutant, grade 2–4; oligodendroglioma, IDH mutation with 1p/19q co-deletion; and glioblastoma, IDH wild type, which is sufficient to see the important role of IDH in glioma.

* Corresponding author.

E-mail address: wj.qin@siat.ac.cn (W. Qin).

¹ Xiang Liu and Wanming Hu and Songhui Diao contributed equally to this work.

Since IDH gene mutations are extremely important for gliomas, it is crucial to detect whether a patient has a mutation in this gene. Currently, in clinical practice, the common methods for the detection of IDH mutations are based on immunohistochemistry (using IDH1-R132H antibody) or gene sequencing [7]. Nonetheless, both approaches come with certain drawbacks. One of these drawbacks is that immunohistochemistry can solely detect the more prevalent IDH1R132H mutation samples. Additionally, less frequent IDH mutations and other amino acid mutations at the IDH1R132 site necessitate supplementary molecular techniques, which are both time-intensive and demanding in terms of labor. Furthermore, the sensitivity of gene sequencing in the current standard mutation screening is limited [8]. Overall, immunohistochemical methods can suffer from underdiagnosis. Next generation sequencing (NGS) is difficult to implement, expensive, time-consuming, and not widely available. Currently, most hospitals, especially primary hospitals, can only label a patient's pathological diagnosis as "NOS (Not Otherwise Specified)," while the accurate pathological diagnosis is the cornerstone of subsequent precise treatment. Therefore, there is an urgent need for a rapid, inexpensive, easy-to-use, comprehensive and accurate IDH mutation detection tool.

Hematoxylin and Eosin (H&E) stained histopathological images are a valuable tool for sophisticated oncology and can be used to assist in the diagnosis of neuroglioma and other tumors [9]. However, pathologists are unable to determine the status of particular genes through visual interpretation of H&E-stained slides, as this may be beyond the ability of human vision to examine. Several recent studies have shown that the effectiveness of deep learning for mutation detection on H&E-stained histopathological images has been demonstrated for certain cancer types. Loeffler et al. deep learning network was able to predict changes in tumor genes and pathways by analyzing routine H&E-stained tissue sections from 23 tumor types, demonstrating that genetic alterations can be predicted from histological images [11]. Kather et al. demonstrated that deep learning could predict microsatellite instability (MSI) directly from histopathology H&E images of gastrointestinal cancer [12]. Jiang et al. used a deep learning model to characterize colorectal cancer patients by histopathological features and achieved an area under the curve of 0.75 for the MYH9, TP53 genes [13].

Deep learning has been validated for the identification of mutated

genes in a variety of tumors, but the identification of IDH mutations in digital gliomas pathology images has been less studied. Momeni et al. used the deep recurrent attention model to predict the mutation state of IDH based on histopathological images of glioma. After learning and training, this deep recurrent neural network focused on the most relevant region in the large input patch by means of attention, and finally achieved 79 % accuracy and 0.86 AUC area under the curve [14]. Liu et al. used a network based on Generative Adversarial Network (GAN) data enhancement method by converting the discrete distribution of training samples into a new continuous distribution and generating synthetic samples with high fidelity based on the estimated new sample distribution. The accuracy of IDH mutation identification was improved from the original 0.794 to 0.853 [15]. Cui et al., using the multi-instance learning strategy, obtained 78.59 % identification accuracy of IDH mutation by convolutional neural network with an attention layer [10]. These three papers have demonstrated the effectiveness of identifying IDH mutations using deep learning for histopathological H&E images of gliomas. Therefore, utilizing deep learning to predict IDH mutations can offer new approaches.

However, currently methods for identification of IDH mentioned above are used only single scale image, whereas histopathology images naturally have multi-scale information. Therefore, our research aims to investigate whether leveraging this multi-scale information can further enhance the accuracy of distinguishing IDH1 gene mutations in glioma tissue pathology images. In order to more intuitively feel the attention field of the neural network to the image under different scale, we used Gradient-weighted Class Activation Mapping (Grad-CAM [16]) to visualize images at 20x and 40x, respectively, as shown in Fig. 1. The 40x image and 20x image contain different scale information. We performed center cropping on the 20x image to obtain the same scale information as the 40x image. Through visualization results, it is observed that deep learning shows different focus information under different magnification images. Therefore, there is potential to further improve the accuracy of identifying IDH1 gene mutations in pathological images of gliomas by fusing the feature information contained in multi-scale images, thus providing an affordable and accurate prediction method for routine clinical diagnosis.

Currently, multiscale strategies have been widely applied in

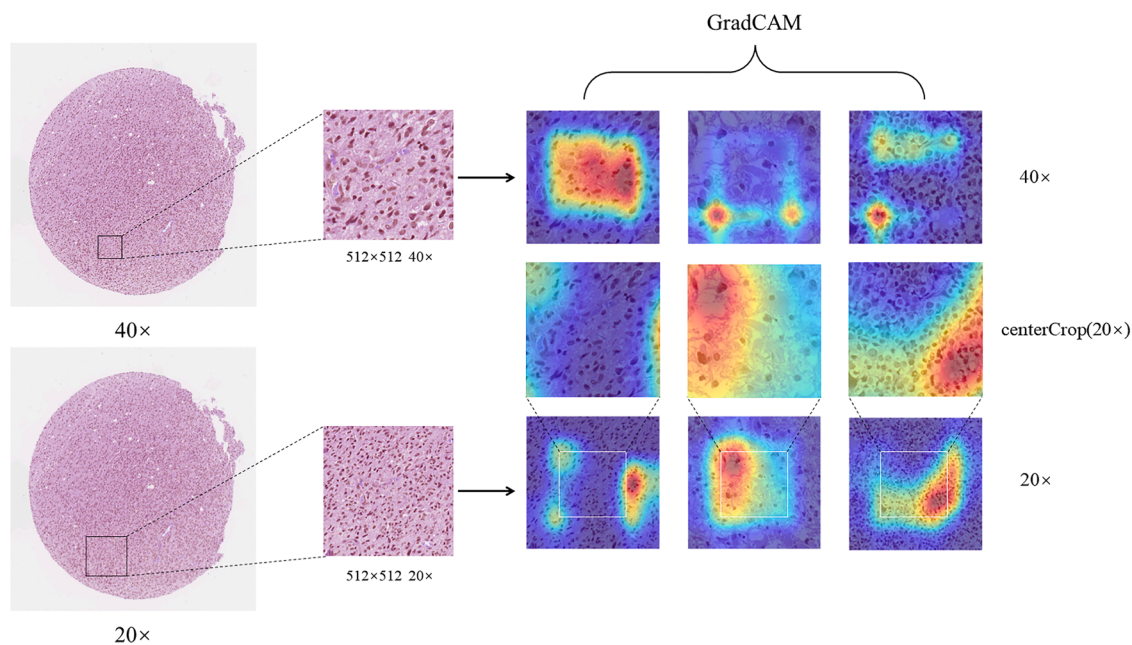


Fig. 1. Visualization of multi-scale information images using GradCAM. Where the darker color proves that the network model pays more attention to it. The first row is the visualized image of the 40x image, the third row is the visualized image of the 20x image, and the second row is obtained by center cropping the visualized image obtained at 20x. The cropped image has the same scale information as the 40x image, so that the comparison is more intuitive.

histopathological images. Liechty et al. used multi-scale images and ensemble learning methods to train four DenseNet121 single-scale models, and predicted IDH mutation with an AUC of 0.87 in brain glioma pathological images [17]. Diao et al. proposed a weak surveillance framework based on multi-scale attention convolutional neural networks (MSAN—CNN) to assist pathologists in efficiently and accurately detecting cancerous regions in whole-slide images of hepatocellular carcinoma [18]. Furthermore, they also proposed a novel deep multi-magnification similarity learning method for precise classification of breast cancer histopathological images [19]. HookNet employed two encoder-decoder networks to extract informative features from two image resolutions and segment lung and breast cancers [20].

Although multi-scale strategies have achieved remarkable results in classification and segmentation tasks of digital pathology, gene mutation prediction tasks based on pathological images are more challenging than tasks such as cancer classification. Because gene mutation prediction is a fine-grained classification task, even experienced pathologists cannot distinguish their differences by naked eye observation. In addition, the multi-scale feature fusion methods currently used in these classification and segmentation tasks often simply merging them together, using simple addition, channel concatenation, and some even resort to training multiple single-scale networks and averaging their results to utilize multi-scale information. These approaches overlook the complex relationships between different scales, which can lead to information loss and confusion. They are unable to accurately differentiate and utilize features at different scales, especially in complex and noisy scenarios. Therefore, directly transferring these methods to gene mutation prediction tasks cannot obtain satisfactory results.

To solve the above problems, we propose a multi-scale fusion gene identification framework (MultiGeneNet). The framework first extracts feature maps from images of different scales through two feature extractors. Then, we introduce a bilinear pooling layer based on the Hadamard product to fuse multi-scale features, which has been proven to be very effective in fusing feature information in fine-grained image classification tasks [23,24]. Finally, we use a discriminator based on more comprehensive and rich multi-scale features to determine whether glioma images have IDH1 genotype mutations.

The contributions of this work can be summarized in the following:

1. We have demonstrated the effectiveness of utilizing deep learning techniques to automatically extract context features from brain glioma pathological images, enabling accurate detection of IDH1 gene mutations.
2. We have demonstrated that the proposed fusion of multi-scale features can enhance the identification of IDH genotypes in digital pathology images of glioma.
3. The proposed MultiGeneNet significantly outperforms other single-scale methods in the evaluation of the H&E stained histopathological tissue microarray (TMA) dataset of brain gliomas under the same experimental conditions.

2. Materials and methodology

2.1. DataSet and data pre-processing

2.1.1. Dataset

In this paper, we employed a glioma TMA dataset consisting of 315 patients with glioma collected approved by the institutional research ethics committee from Sun Yat-sen University Cancer Center, Guangzhou, China. The dataset was designed to establish a link between gene mutations and pathologic images. Glioma samples were collected as follows: first, we marked representative tumor areas and used a tissue arrayer to punch out three cylinders of 1.0 mm diameter tissue from these areas, which were then re-embedded into the recipient paraffin blocks at defined locations. Subsequently, we mounted sections of these TMA blocks on microscope slides for scanning. During sample selection,

we excluded samples from patients with unavailable resected tissue, inaccurate follow-up data, and those who died from diseases unrelated to glioma, resulting in 315 samples from patients with glioma. These samples were bi-directionally validated by immunohistochemistry and sequencing [7] with clear IDH mutation results. The labels provided in the dataset include only IDH1 genotype mutation types, but not other less common IDH mutation types. The dataset spans from 2008 to 2016, and patients were followed for a median of 54 months (range 0–176 months). The unlabeled cases and cases with poor image quality were excluded, and 296 patient samples were finally enrolled in study analysis. These samples consisted of patients in WHO class 2 (89 patients), WHO class 3 (80 patients), and WHO class 4 (127 patients), spanning the age range of 2 to 78 years, with 176 males and 120 females. Additional details of the dataset are provided in Table 1, where SD represents standard deviation.

To enhance the model performance evaluation’s accuracy, we selected an independent test set comprising 36 % of the dataset, ensuring that these data points were entirely excluded from the training process. Moreover, we based our selection of images on individual patients rather than individual images. This deliberate approach ensured that images from the same patient did not appear in both the training and test sets, effectively mitigating the risk of information leakage and enhancing prediction accuracy. In addition, we also calculated the p-value of WHO, IDH-type, Gender, and Age variables between the two groups of the training and test sets. Notably, all p-values exceeded 0.05, providing evidence that there were no significant differences in age, gender, and other variables between the two groups we divided for training and testing.

2.1.2. Data pre-processing

Due to the differences in raw materials and manufacturing techniques of stain suppliers, the dose of stain placed between different laboratories, and the time of slide placement, pathology images are very prone to varying shades of color. Since deep learning is a very sensitive model for color information [35], a color normalization is usually performed on pathology images. The color normalization algorithm used in this paper is the Vahadane algorithm [21]. The Vahadane algorithm uses an unsupervised strategy to decompose the image into sparse and non-negative staining density maps, which are combined with template images for the purpose of normalizing staining and preserving the tissue structure of the pathology image.

Due to the high resolution of the images in the dataset, we have chosen to address computational efficiency and storage challenges by implementing a strategy of segmenting the large images into smaller patches. In this study, we divide the original image into 1024×1024 patches by employing overlapping cuts with a step size of 512. This approach not only increases the dataset’s sample size but also mitigates any disruption to the cellular structure caused by splitting at the cut boundaries. During the process of cutting the patches, we eliminate the patches with blank areas larger than 70 %. Additionally, in order to avoid potential biases that may arise due to the different number of

Table 1
Dataset details.

Characteristics	Train(N = 190)	Test(N = 106)	Total(N = 296)	P-value
WHO				0.869
WHO(2)	59 (31.0 %)	30 (28.3 %)	89 (30.0 %)	
WHO(3)	50 (26.3 %)	30 (28.3 %)	80 (27.0 %)	
WHO(4)	81 (42.7 %)	46 (43.4 %)	127 (43.0 %)	
IDH-type				0.948
wildtype	109 (57.4 %)	62 (58.5 %)	171 (57.7 %)	
mutant	81 (42.6 %)	44 (41.5 %)	125 (42.3 %)	
Gender				0.110
Male	106 (55.8 %)	70 (66.0 %)	176 (59.4 %)	
Female	84 (44.2 %)	36 (34.0 %)	120 (40.6 %)	
Age(mean±SD)	42.0 ± 16.4	42.6 ± 15.1	42.2 ± 15.97	0.768

patches in an image, we uniformly reserve 25 patches for each original image.

To prevent overfitting problems and improve the model's robustness, we performed data augmentation on the dataset [22]. We employed two operations in the data augmentation process: image flipping and enhancement of image color, contrast, and brightness. These operations aimed to expand the dataset while preserving the essential information within the original HE images. Additionally, specific quantitative details are provided below to facilitate reproducibility in future research. Firstly, image flipping was achieved using the `transpose(Image.FLIP_LEFT_RIGHT)` function from the Python Image library. This operation effectively increases the number of samples in the dataset and does not change the basic shape of the cells. Secondly, the PIL library's Image-Enhance module was utilized to enhance the image's color, contrast, and brightness. Specifically, three random numbers ranging from 0.8 to 1.2 were generated and applied to Color, Contrast, and Brightness functions to transform the image. Through this approach, we were able to simulate realistic variations observed in actual images while preserving the histopathological content.

2.2. Workflow overview

Fig. 2 demonstrates the entire data processing flow and the details of MultiGeneNet. Firstly, the images are extracted from the dataset and loaded with labels. Several data preprocessing techniques are then performed, followed by data augmentation techniques to increase the size of the dataset. Finally, the dataset is divided into a training set and a test set. We train the model with a private digital pathology image dataset of glioma and evaluate MultiGeneNet using an independent test set. The details of each component are described in the following

subsections.

2.3. Bilinear pooling layer

The bilinear pooling layer was first proposed in 2015 by Lin et al. [23]. In this paper, two feature extractors are used, and then the fused feature maps are obtained by using external product multiplication at each position of the feature images output from the two feature extractors. This feature fusion strategy proved to be highly effective for fine-grained classification.

We assume that the two feature extractors are $f_A(\cdot)$, $f_B(\cdot)$, and for an image we use two feature extractors to extract the feature map of the image, then the two features of the image at its k positions are $f_A(k, X) \in \mathbb{R}^{T \times M}$ and $f_B(k, X) \in \mathbb{R}^{T \times N}$ (For the familiar image features, $T = 1$, while M and N are the number of feature channels, i.e., the number of feature maps), then the fusion operation of features for image X using a bilinear pooling layer is:

$$b(k, X, f_A, f_B) = f_A^T(k, X) f_B(k, X) \in \mathbb{R}^{M \times N}$$

$$\delta(X) = \sum_k b(k, X, f_A, f_B) \in \mathbb{R}^{M \times N}$$

$$x = \text{vec}(\delta(X)) \in \mathbb{R}^{MN \times 1}$$

$$y = \text{sign}(x) \sqrt{|x|} \in \mathbb{R}^{MN \times 1}$$

$$z = y / \|y\|_2 \in \mathbb{R}^{MN \times 1}$$

(1)

The above formula first performs a fusion operation (actually a multiplication operation) at the same positions of the feature maps

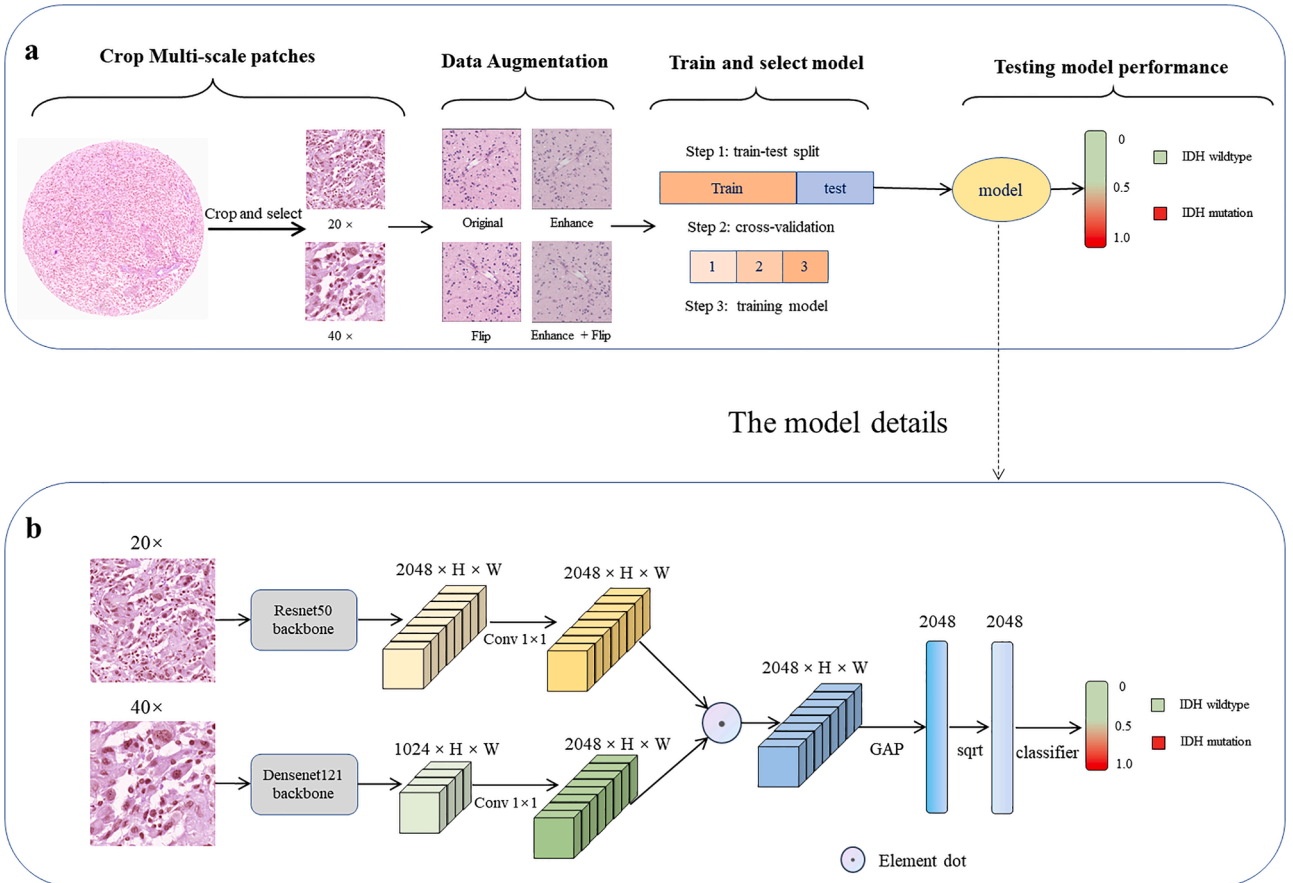


Fig. 2. a: Data processing flow b: MultiGeneNet details, conv1x1 is two-dimensional convolution operation and the convolution kernel size is 1 × 1, GAP is the global average pooling operation, sqrt is the square root of the values of the elements and sign unchanged, classifier is a fully connected layer.

extracted by the two feature extractors, and then an $M \times N$ matrix is obtained in this way. Then the fused feature matrix b at each position of the feature map is calculated, and the feature fusion matrix δ of image X is obtained by the summation operation. Then this matrix is transformed into a tensor, which is noted as a bilinear vector x . Then the unsigned open root operation and the L2 normalization operation are performed on it to obtain the final fused features, and finally the fused features z can be used for subsequent classification operations.

For image features, the above formula can be further simplified. Suppose $a_k = f_A^T(k, X) \in \mathbb{R}^{M \times 1}$, $b_k = f_B^T(k, X) \in \mathbb{R}^{N \times 1}$, then the original matrix $\delta X = \sum_k a_k b_k^T \in \mathbb{R}^{M \times N}$.

Suppose $A = [a_1, a_2, \dots, a_k] \in \mathbb{R}^{M \times K}$, $B = [b_1, b_2, \dots, b_k] \in \mathbb{R}^{N \times K}$, then we can completely use matrix multiplication to replace the cumulative symbol $\delta X = AB^T \in \mathbb{R}^{M \times N}$. Thus, the above equation simplifies to:

$$\delta(X) = AB^T \in \mathbb{R}^{M \times N}$$

$$x = \text{vec}(\delta(X)) \in \mathbb{R}^{MN \times 1}$$

$$y = \text{sign}(x) \sqrt{|x|} \in \mathbb{R}^{MN \times 1}$$

$$z = y / \|y\|_2 \in \mathbb{R}^{MN \times 1} \quad (2)$$

However, the bilinear pooling layer exhibits a notable drawback: it generates an excessive number of fused features. For instance, if both CNN feature extractors each produce 1000 feature maps, the resulting fused feature maps would amount to $1000 \times 1000 = 1,000,000$. In the original paper, the authors attempted to mitigate this issue by employing PCA to reduce feature dimensionality, but the final outcome was less than satisfactory. Consequently, many researchers have since sought to enhance the original bilinear pooling layer.

2.4. Multi-scale fusion gene identification network

Fig. 2b shows the model details of MultiGeneNet, which combines two popular pre-trained models, ResNet50 and DenseNet121, to create a multi-scale fusion gene identification framework. The two models extract feature information at two scales from the brain glioma histological images at 20x and 40x, respectively. Then, they employ a bilinear pooling layer based on the Hadamard product to merge the features extracted at two scales. Finally, they use the full connection layer to integrate the fused features and perform the classification task. The basic structure of each model uses a pre-trained CNN model.

2.4.1. Factorized bilinear pooling based on Hadamard product

Assume that the feature map matrix of an image I after automatic feature extraction by a feature extractor is $X \in \mathbb{R}^{c \times h \times w}$, where h and w represent the height and the width of the feature map, respectively, and c represents the number of channels, i.e., the number of feature maps. Assume that the feature vector $x = [x_1, x_2, \dots, x_c]^T$, inspired by [24]. To better utilize spatial location information, the feature vector is transformed as follows:

$$z_i = x^T W_i x \quad (3)$$

where z_i is the output value of the bilinear model, which is the fused eigenvalue, and $W_i \in \mathbb{R}^{c \times c}$ is the projection matrix, which is the parameter we need to learn. According to matrix decomposition, the projection matrix can be decomposed into two one-dimensional vectors as follows:

$$z_i = x^T W_i x = x^T U_i V_i^T x = U_i^T x \circ V_i^T x \quad (4)$$

where $U_i \in \mathbb{R}^c$, $V_i \in \mathbb{R}^c$. Thus, the fused feature vector $z \in \mathbb{R}^d$ is defined as

$$z = U^T x \circ V^T x \quad (5)$$

where $U \in \mathbb{R}^{c \times d}$, $V \in \mathbb{R}^{c \times d}$, $\circ$ is the element corresponding to the product operation, and d is the length of the fused feature vector. The final fused feature map matrix $Z \in \mathbb{R}^{d \times h \times w}$ is obtained by computing the feature vector for each spatial location of the feature map matrix X .

2.4.2. Multi-scale fusion for gene identification

The bilinear pooling layer offers the capability to merge feature maps obtained from two distinct feature extractors, leading to improved classification accuracy, particularly in fine-grained image classification tasks. In this study, we focus on the analysis of histopathological images to discern the IDH1 mutation status. Given the current limitations where even skilled pathologists struggle to confirm mutation status through visual pattern analysis alone, we recognize this as a fine-grained classification challenge that surpasses human visual inspection capabilities. Hence, our approach involves employing bilinear pooling layers for Hadamard products to combine feature maps encompassing diverse scale information. Subsequently, we carry out the classification task using the fused multi-scale features.

We assume that the two feature extractors ResNet50 and DenseNet121 are $f_A(\cdot)$ and $f_B(\cdot)$, and the feature map matrix extracted from the $20 \times$ image I_{20} by the feature extractor ResNet50, $X = f_A(I_{20}) \in \mathbb{R}^{c \times h \times w}$, and the feature map matrix extracted from the $40 \times$ image I_{40} by the feature extractor DenseNet121, $Y = f_B(I_{40}) \in \mathbb{R}^{c \times h \times w}$, then the fused multi-scale feature map matrix Z is defined as

$$Z = U^T X \circ V^T Y \quad (6)$$

where $U \in \mathbb{R}^{c \times d}$, $V \in \mathbb{R}^{c \times d}$, $Z \in \mathbb{R}^{d \times h \times w}$. The final output $O \in \mathbb{R}^{l \times 1}$ is defined as

$$\delta(X, Y) = \text{GAP}(Z) \in \mathbb{R}^{d \times 1 \times 1}$$

$$v_1 = \text{vec}(\delta(X, Y)) \in \mathbb{R}^{d \times 1}$$

$$v_2 = \text{sign}(v_1) \sqrt{|v_1|} \in \mathbb{R}^{d \times 1}$$

$$O = P^T v_2 \in \mathbb{R}^{l \times 1} \quad (7)$$

GAP is the global average pooling operation, $P \in \mathbb{R}^{d \times l}$ is the classification matrix, which is actually the parameters of the fully connected layer, where l represents the number of categories in the classification task. The overall framework diagram is shown in Fig. 2b.

2.4.3. Loss function and calculates the original image label

We assume the model is a function f_θ with current parameter θ that maps input patches to class probabilities for wildtype and mutant type. All selected patches from an image share the same target y_i , the loss can be computed with Eq. (8) for each of the patches:

$$L = \frac{1}{N} \sum_{i=1}^N - [y_i \cdot \log(\tilde{y}_i) + (1 - y_i) \cdot \log(1 - \tilde{y}_i)] \quad (8)$$

where L is the loss value, N is the number of samples, and $\tilde{y}_i = f_\theta(I_{20}, I_{40})$ represents the prediction probability of a patch. The parameters of the model are adjusted by minimizing the cross-entropy loss function, so that the prediction results of the model are closer to the real label.

Since our model only acquires prediction results of patches, for an original image, the model will generate a probability array $\tilde{Y} = \{\tilde{y}_i : i = 1, 2, \dots, n\}$. In order to obtain the label of the original image, we use the voting strategy, taking 0.5 as the threshold. Finally, the prediction result of the original image can be calculated using the Eq. (9):

$$Y_{\text{pred}} = \begin{cases} 1 & \text{if } \text{number}(\tilde{Y} > 0.5) > \text{number}(\tilde{Y})/2 \\ 0 & \text{else} \end{cases} \quad (9)$$

where 1 represents the mutation type and 0 represents the wild type. The

number function is used to calculate the number of elements in the array that satisfy the condition.

3. Experimental results

3.1. Training details and experimental metrics

To prevent overfitting, we utilized the principle of transfer learning [25] and used the training parameters from the ImageNet dataset. This dataset has a large amount of data and sufficient categories, which allowed us to use it as the initialization parameters for the feature extractor. By doing this, we ensured that the network had better feature extraction capability and improved model robustness. In addition, for the selection of some hyper-parameters, we used the Adam optimizer and the cross-entropy loss function, and the learning rate was set to 0.0001.

To measure the performance of the model, we chose the following measures: Precision, Recall, Accuracy, F1-score, and area under the curve (AUC), which depend on the calculation results of the confusion matrix, including true positive (TP), true negative (TN), false positive (FP), and false negative (FN).

These measures were calculated as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (10)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (11)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + TN + FP} \quad (13)$$

The final experimental outcome represents the validation of the model's performance based on the optimal classification accuracy achieved through a 3-fold cross-validation approach on an independent test set. We avoided employing a 5-fold cross-validation primarily due to the limited number of patients in the dataset. Our approach relied on selecting the final model based on the best classification accuracy from the validation set. Choosing a model based on just one-fifth of the training data might lead to insufficient robustness and generalization, potentially resulting in strong performance on a small validation set but poor results on an independent test set. Consequently, we adopted a 3-fold cross-validation strategy to enlarge the validation set for each fold, ensuring a more generalized selection of the final model.

3.2. MultiGeneNet compared to the single scale model

Table 2 shows the results of a comparison between MultiGeneNet and six other single-scale models of transfer learning. MultiGeneNet extracted feature information at 20× and 40× using ResNet50 and DenseNet121, respectively, and then used the fused multiscale features to identify IDH1 mutations in glioma histopathology images. In contrast,

Table 2

Classification performance of MultiGeneNet and other single-scale models on the test set. The best results are shown in bold.

Models	Precision	Recall	F1-score	Accuracy	AUC
MobileNetV3	0.6556	0.6860	0.6705	71.9807	0.806
Resnet50	0.7805	0.7442	0.7619	80.6763	0.858
Densenet121	0.7363	0.7791	0.7571	79.2271	0.855
Res2net50	0.8082	0.6860	0.7421	80.1932	0.874
Vgg16	0.8082	0.6860	0.7421	80.1932	0.822
Inception_v3	0.7971	0.6395	0.7097	78.2609	0.831
Efficientnet-b7	0.7215	0.6628	0.6909	75.3623	0.827
MultiGeneNet	0.8333	0.7558	0.7927	83.5749	0.886

the other six single-scale models only utilized 20x pathology images. In this paper, we consider 20x images as the primary input image and 40x images as auxiliary information for two reasons: first, 20x images have more visual field information than 40x images at the same resolution; second, we investigated other studies based on histopathology images for gene identification [26,27,28], and a large number of authors indicated that the models trained with 20x images have better classification performance. We speculate that this might be because higher magnification images, such as 40x, cause the model to focus solely on the details of cells, disregarding the spatial connections between cells and other features. On the other hand, lower magnification images, such as 10x and 5x, have insufficient information to offer, and 20x images take into account the image quality and larger field of view, resulting in better classification performance.

Based on the results, MultiGeneNet achieves the best performance in terms of Precision, F1-score, Accuracy, and AUC. The AUC value is 0.886, which proves that MultiGeneNet has better classification stability than other models. Additionally, the Accuracy is 83.5749 %, which is nearly 3 % higher than other models. Moreover, the Precision reaches 0.8333, which is better than other transfer learning models. However, the Recall result is lower than that of the Densenet121 transfer learning model. It is worth noting that Precision and Recall sometimes contradict each other. For instance, Densenet121 is better than MultiGeneNet in Recall but worse than MultiGeneNet in Precision, which makes it challenging to judge the pros and cons of the models. Therefore, we use the comprehensive index F1-score of the two for measurement, and the F1-score of MultiGeneNet reaches 0.7927, which is significantly better than other models. In conclusion, our MultiGeneNet's effectiveness in identifying IDH1 mutations in brain glioma histopathological images is significantly better than that of other single-scale methods.

3.3. Ablation studies of multigenenet

We conducted ablation experiments using the MultiGeneNet model to gain more insight into the performance impact associated with the different modules of our model. Table 3 shows the results of our ablation experiments. Since MultiGeneNet is based on 20x and 40x images as input and fuses two feature maps containing different scale information, we explored only input 20x, only input 40x and fusion strategies of the multi-scale features respectively. Among them, the trained model with only 20x or only 40x input lags behind MultiGeneNet in terms of precision, F1-score, accuracy, and AUC. This proves that the complementarity of information between images with different scales is helpful in improving the identification effect of IDH1 mutation. Furthermore, the information presented in Table 3 indicates that combining multi-scale features through channel concatenation does not enhance the classification performance; instead, it diminishes it. We believe that this concatenation method merely expands the number of features, thus adding many useless or even inhibiting features to the classification effect, and finally leading to the deterioration of the classification effect. The multi-scale fusion strategy in MultiGeneNet multiplies features of different scales by specific weights, which are automatically learned through the network. Throughout the learning process, the weights assigned to irrelevant features are diminished, while those associated with valuable features are augmented. This ultimately enhances the overall classification performance. Based on these ablation experiments, we concluded that effective fusion of multi-scale information can improve the identification of IDH1 mutations.

3.4. Analysis of proposed multigenenet

Fig. 3 displays the confusion matrix and the Receiver Operating Characteristic (ROC) for ResNet50, the channel-concatenated multiscale fusion strategy, and MultiGeneNet. The test dataset comprises 207 original images (1 or 2 images per patient) from 106 patients, as presented in Table 1. Upon examining the visualized outcomes, it is evident

Table 3
Ablation experiment of MultiGeneNet. The ablation was performed based on the 20x, 40x, and fusion strategies: The "concat" operation represents the stacking of features on the channel. The results in the last row represent the classification performance of MultiGeneNet, with the best results shown in bold.

20×	40×	Fusion strategy	Precision	Recall	F1-score	Accuracy	AUC
✓			0.7805	0.7442	0.7619	80.6763	0.858
	✓		0.7444	0.7791	0.7614	79.7101	0.854
✓	✓	Concat	0.8082	0.6860	0.7421	80.1932	0.855
✓	✓	Bilinear Pool	0.8333	0.7558	0.7927	83.5749	0.886

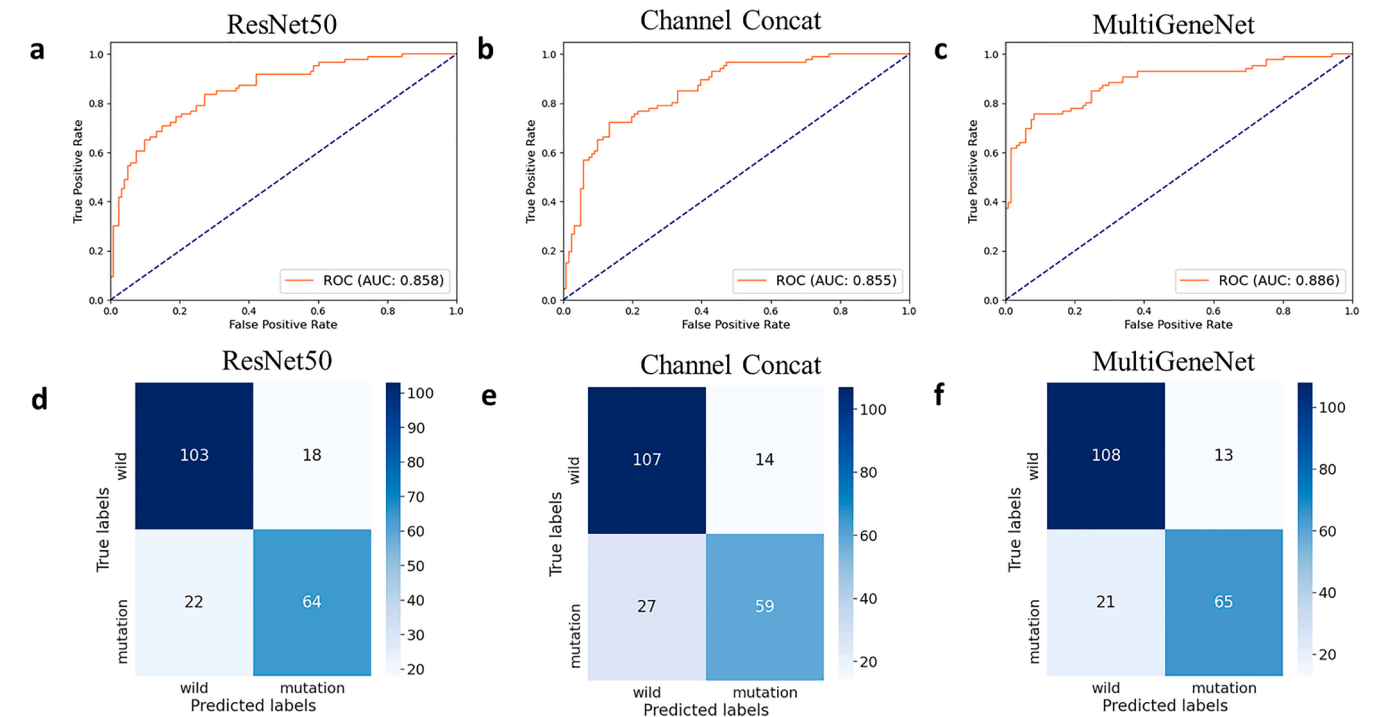


Fig. 3. Confusion matrix and ROC curve drawn from the evaluation results of ResNet50, channel fusion strategy and MultiGeneNet on the test set.

that MultiGeneNet outperformed the other methods, achieving the highest prediction performance with an AUC metric of 0.886. This substantiates that MultiGeneNet exhibits superior robustness and generalization compared to the alternative models. From our examination of the confusion matrix results, it's evident that the IDH1 mutant and IDH1 wild type categories were accurately classified in 65 and 108 images, respectively. Notably, the number of correctly predicted samples for each category exceeded those achieved by both the ResNet50 single-scale method and the multi-scale fusion approach using channel concatenation. Despite the channel concatenation strategy utilizing all the 20x and 40x image features, this simplistic concatenation method introduces numerous redundant and uninformative features, which adversely impact the final classification outcomes. In contrast, the MultiGeneNet model is capable of attenuating the influence of extraneous feature information by learning appropriate weights, leading to superior classification results.

Furthermore, we noticed that the MultiGeneNet's prediction accuracy for IDH1 mutants exhibited only marginal improvement, and surprisingly, even the accuracy of the channel concatenation approach saw a significant decline after incorporating feature information from the 40x images. This led us to hypothesize that the model may have excessively emphasized the morphological attributes of individual cells at the 40x image scale. It is plausible that the distinctive characteristics of IDH1 mutations primarily manifest through the spatial relationships between cells within histopathological images. Consequently, the fusion of characteristic information at 40x scale predominantly bolstered the accuracy of wild type identification, with only slight gains in mutant

identification accuracy. Nevertheless, taken as a whole, MultiGeneNet substantially enhances the classification efficacy of IDH1 genotypes by effectively amalgamating information from diverse scales.

3.5. MultiGeneNet prediction effect visualization

Fig. 4a illustrates the prediction probabilities of each model in the test set, revealing a noteworthy difference. The multiscale model consistently demonstrates significantly higher prediction probabilities compared to the single-scale model. This advantage arises from the multiscale model's ability to incorporate diverse contextual information at varying scales, providing it with a richer understanding of the input data and subsequently boosting confidence in its predictions.

Furthermore, MultiGeneNet surpasses the concatenation strategy in terms of stability and prediction probabilities. The feature fusion strategy in MultiGeneNet effectively capitalizes on the complementarity between various features, yielding more comprehensive and information-rich feature representations. In contrast, the channel concatenation strategy simply combines features by channel without explicitly modeling their correlations. Consequently, our model excels at capturing the intricate structure and inherent relationships within the input data.

UMAP algorithm is used to visualize the features extracted by MultiGeneNet. As illustrated in Fig. 4b, the majority of the sample features were accurately clustered into their respective groups, thereby demonstrating the existence of unique feature patterns that differentiate mutant from wild-type. The trained MultiGeneNet was able to capture

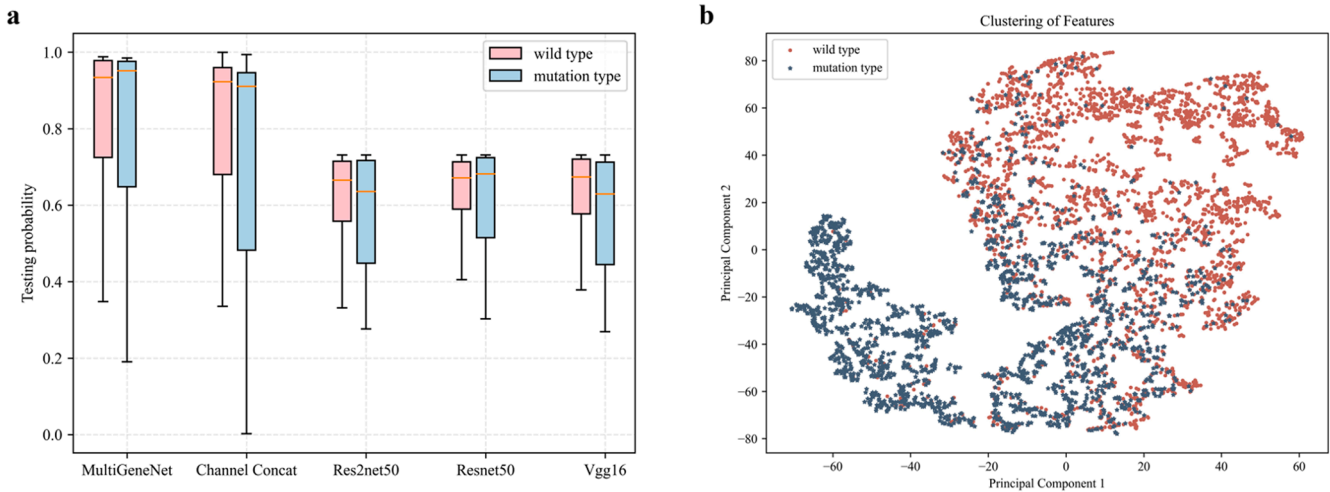


Fig. 4. Visualization of MultiGeneNet mutation identification performance **a:** Visualization of predictive probability of each model on independent test set. **b:** Visualization of feature clustering extracted from MultiGeneNet using UMAP.

the underlying features between different types of samples.

However, it's worth noting that some sample features were erroneously assigned to the wrong clusters in the clustering results. This could be attributed to the cropped patches lacking typical features associated with their label category, thereby confounding the clustering algorithm's efforts to allocate them correctly. This may also contribute to the challenges faced by MultiGeneNet in achieving optimal classification performance. In our future research endeavors, we plan to explore the implementation of a multiple instance learning strategy to further enhance mutation identification.

4. Discussion

4.1. Discussion of ViT model predictions

Currently, research on gene mutation prediction based on pathological images mainly adopts CNN methods [10,14,15,31,32,33,34], while relatively few studies have used vision transformer (ViT) models. However, vision transformer models have made significant progress in natural image classification tasks, which has sparked our curiosity about their performance in gene mutation identification tasks. Therefore, to explore the potential of vision transformer models in gene mutation prediction, we conducted experiments on brain glioma tissue pathological images using three models: NoisyViT [29], ViT-B [30], and ViT-L [30], and the experimental results are summarized in Table 4.

As can be seen from Table 4, the classification performance of the NoisyViT model is poor. Although this model is currently the best performing visual deformation model on the ImageNet dataset, its direct training results are relatively poor because the authors' pre-training model is not publicly available. In contrast, the ViT-B model using the pre-training parameters exhibits comparable or even better classification performance than the CNN model. Even, the classification performance of the ViT-L model with a patches parameter of 32 is significantly

better than that of the CNN model, with an accuracy of 83.0918 %, which is close to the recognition performance of the MultiGeneNet model using CNN. This indicates that the ViT model with pre-trained parameters shows superior performance in the IDH1 recognition task.

Furthermore, we further explored whether a multi-scale information fusion strategy could improve the performance of ViT models. Applying the multi-scale fusion strategy to the ViT-L/32 model successfully increased the classification accuracy to 85.0242 %, further demonstrating the effectiveness and superiority of this strategy.

4.2. Comparison with other methods for gene mutation identification

In the current research, relatively little work has been done on predicting gene mutations based on pathology images, so we have conducted an extensive discussion on the methods in the task of gene mutation identification to provide a comprehensive perspective, including three single-scale methods and one multi-scale method. Detailed experimental results can be found in Table 5. The experimental results show that our proposed MultiGeneNet model performs optimally on several performance metrics. Of particular note, Liechty's multi-scale approach also shows better performance than other single-scale strategies, which demonstrates that fusing multi-scale information does help to improve the discrimination of gene mutation prediction.

In addition, we find that our model outperforms Liechty's multi-scale model. We suppose that this is because Liechty's multi-scale model is too simple and performs information fusion only by averaging the prediction results of multi-scale images. This approach does not adequately consider the connection between multi-scale information, which may lead to information redundancy and confusion, thus affecting the final performance metrics. In contrast, our multi-scale information fusion strategy, which is dynamically tuned by network learning, is able to better consider the connections between multi-scale information and fuse them in a more optimal way.

4.3. Discussion of feature maps in multigenenet

In order to investigate the inference basis of MultiGeneNet in gene mutation identification, we performed the visualization of the middle feature layer and presented the results in Fig. 5. The observations revealed that the shallow features exhibit a high degree of compatibility with the cell outline information, indicating that the network starts to capture the fundamental structural and morphological details of the image during the early learning stages. Additionally, the deep features demonstrate a stronger focus on areas with clustered cells exhibiting

Table 4

The performance effect in different Vit models and the application of our multi-scale information fusion strategy on ViT models.

Models	Precision	Recall	F1-score	Accuracy	AUC
NoisyViT	0.6974	0.6163	0.6543	72.9469	0.771
ViT-B/16	0.8026	0.7093	0.7531	80.6763	0.843
ViT-B/32	0.8169	0.6744	0.7389	80.1932	0.874
ViT-L/16	0.8101	0.7442	0.7758	82.1256	0.869
ViT-L/32	0.8000	0.7907	0.7953	83.0918	0.886
ViT-L/32 + Multi scale	0.8089	0.8372	0.8228	85.0242	0.891

Table 5
Classification performance of MultiGeneNet and other gene mutation identification models on the test set. The best results are shown in bold.

Author	Scale strategy	Ref.	Precision	Recall	F1-score	Accuracy	AUC
Kather	Single scale	[31]	0.7821	0.7093	0.7439	79.7101	0.853
Kather	Single scale	[32]	0.7701	0.7791	0.7746	81.1594	0.872
Fu	Single scale	[33]	0.7975	0.7326	0.7636	81.1594	0.873
Liechty	Multi scale	[34]	0.8356	0.7093	0.7673	82.1256	0.870
Our	Multi scale		0.8333	0.7558	0.7927	83.5749	0.886

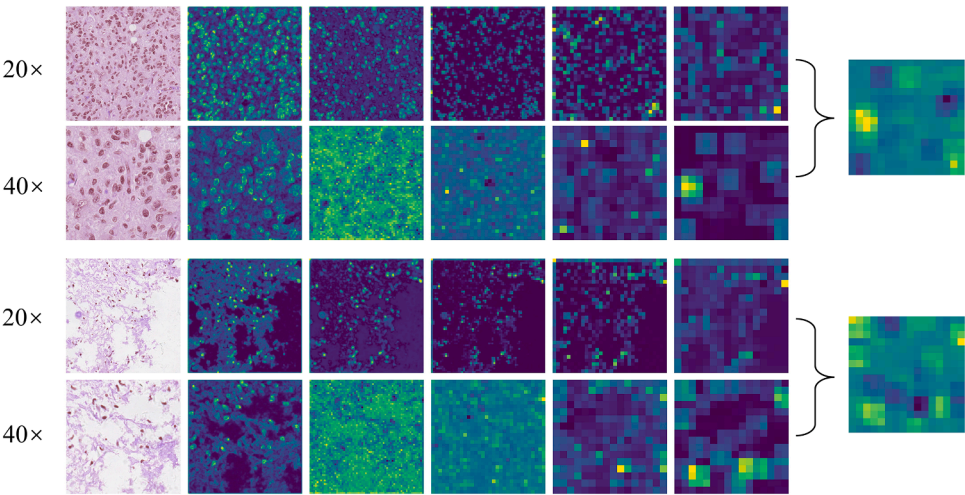


Fig. 5. Visualization of the middle feature layer extracted by MultiGeneNet, with deeper depths from left to right. The rightmost column is the fused multi-scale feature map.

morphological changes and spatial connections, while paying minimal attention to the background regions. These results provide compelling evidence that the network relies on pathologically relevant features in the images to accurately identify gene mutations.

In addition, we observed that the fused multi-scale features exhibit a more comprehensive and representative feature representation compared to single-scale features. This indicates that our fusion strategy effectively utilizes multi-scale features, resulting in superior discriminative performance compared to single-scale features. This further validates the effectiveness of our multi-scale fusion strategy.

5. Conclusion and limitations

In the current study, deep learning techniques have been shown to be useful for predicting IDH1 mutation status in histopathologic images of brain gliomas [10,14,15,17]. However, these methods use only single-scale image features and ignore the multi-scale information inherent in pathological images. Therefore, we propose MultiGeneNet, which uses bilinear pooling layers based on Hadamard product to fuse feature information, and uses network parameter learning to reduce the influence of redundant useless features, effectively capturing the interaction between different features. Our experimental results show that MultiGeneNet has significant advantages in predicting the mutation status of IDH1 gene. On an independent test set, our method achieves an AUC value of 0.886 and an accuracy of 83.5749 %, which is significantly better than other single-scale methods. Through ablation experiments and box plot analysis, we further validated the effectiveness and superiority of the multi-scale fusion strategy.

In addition, we discuss the performance of the ViT model in predicting IDH1 gene mutations in glioma pathology images. The experimental results show that the performance of the ViT-L model with pre-training parameters is significantly better than that of the CNN model. In particular, the classification accuracy of the model is improved from 83.0918 % to 85.0242 % after introducing our proposed multi-scale

information fusion strategy. This result further confirms the effectiveness of the multi-scale information fusion strategy and demonstrates its generalization across different types of models, both CNN and Transformer models. Finally, we further validated the novelty and effectiveness of our approach through feature map visualization and comparison with other existing methods. The visualization of these intermediate feature layers also reflects the network’s ability to capture relevant structural and morphological details, enabling accurate identification of gene mutations based on pathological features.

Our study demonstrates the excellent performance of deep learning models in identifying gene mutations, however, we acknowledge that there are limitations to our research. Firstly, we are limited to a specific type of brain glioma tissue pathology images, thus it is necessary to expand the dataset to verify the model’s generalizability to other types of pathological images. Secondly, our study only involves data from a specific medical institution, and further verification of the model’s robustness and reliability on larger and more diverse datasets is required. Additionally, we only explored the fusion of 20x magnification information and 40x magnification information, future studies can explore the fusion of additional magnification levels to investigate whether it can further enhance the recognition of IDH mutations. Finally, our study has not yet to been applied in clinical hospitals. Future larger clinical trials are needed to verify the accuracy and reliability of the method, and comparisons with standard diagnostic methods should be conducted.

Despite these limitations, our research suggests that deep learning models have potential and effectiveness in predicting gene mutations. Future studies can further explore the fusion of multiple magnification levels of information, and combine biological knowledge and prior information to guide model training, in order to establish more comprehensive predictive models. Additionally, applying the feature representation of deep learning models to explain the biological mechanisms of gene mutations is also an interesting research direction, which will contribute to the development and progress of medical image

analysis.

In conclusion, MultiGeneNet leverages multi-scale feature fusion to map histopathological images to IDH1 gene expression and achieves strong prediction results. Currently, IDH mutations are primarily detected through gene sequencing and immunohistochemistry, as visual examination of H&E stained histological specimens is insufficient for pathologists to determine IDH mutation status. Moreover, the availability of medical resources varies significantly among countries, making it impossible to deploy technologies like immunohistochemistry and gene sequencing in underdeveloped or resource-limited areas, such as certain regions in Africa. Therefore, MultiGeneNet, as an artificial intelligence technology, offers a cost-effective and compatible solution for these regions, serving as a diagnostic or prognostic tool.

Statement of ethical approval

Ethical approval has been granted for the collection and use of the data analyzed in this study. The approval number is: B2021–178-Y01.

CRediT authorship contribution statement

Xiang Liu: Conceptualization, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft. **Wanming Hu:** Data curation, Resources, Writing – review & editing. **Songhui Diao:** Methodology, Software. **Deboch Eyob Abera:** Writing – review & editing. **Daniel Racocanu:** Investigation, Writing – review & editing. **Wenjian Qin:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors affirm that there are no financial interests or personal relationships that could have influenced the work reported in this paper.

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